

Double channel high-side driver

Features

Description	Parameter	Value
Max. transient supply voltage	V_{CC}	52V
Operating voltage range	V_{CC}	8 to 36 V
Typ. on-state resistance (per-ch)	R_{ON}	50 m Ω
Current limitation (typ.)	I_{LM}	20 A
Off-state supply current	I_s	2 μ A

● General

- Very low standby current
- 3.0V CMOS compatible input
- Optimized electromagnetic emission
- Very low electromagnetic susceptibility
- Improved system level reliability
- Fault status reset pin (FR)

● Diagnostic functions

- Output short to VCC detection
- Overload and short to ground latch-off
- Thermal shutdown latch-off

● Protections

- Under-voltage shutdown
- Overvoltage clamp
- Load current limitation
- Self-limiting of fast thermal transients
- Protection against loss of VCC
- Thermal shutdown
- Electrostatic discharge protection

Applications

Common automotive applications

Bulbs with high inrush current loads

Replace relays and fuses

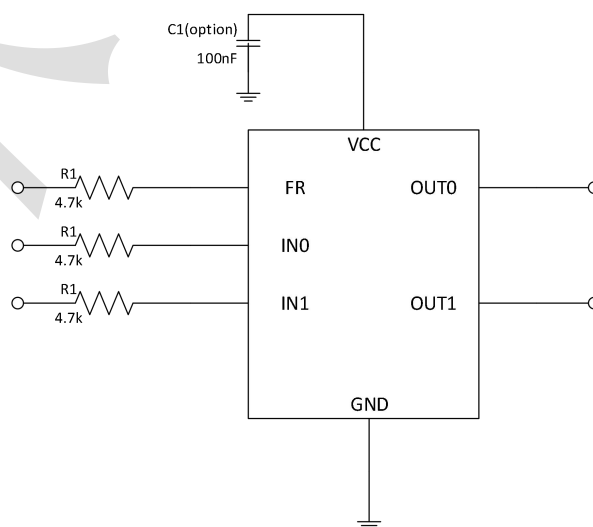
Descriptions

The INH22050 is a monolithic device made using present advanced semiconductor process. The device is intended for driving resistive or capacitive loads with one side connected to the ground. Active V_{CC} pin voltage clamp protects the device against low energy spikes. This device integrates robust protection features to ensure output port protection even during harmful events like short circuits in all kinds of systems.

The high current limit range allows for usage in loads that require large transient currents, while the low current limit range provides improved protection for loads that do not require high peak current. The device is capable of reliably driving various load profiles. Fault conditions such as overload, over-temperature, or short to VCC. Output current limitation protects the device in overload conditions. The device latches off in case of overload or thermal shutdown.

The device is reset by a low level pass on the fault reset pin(FR).

Application Circuit



Block diagram and pin description

Figure 1. Block diagram

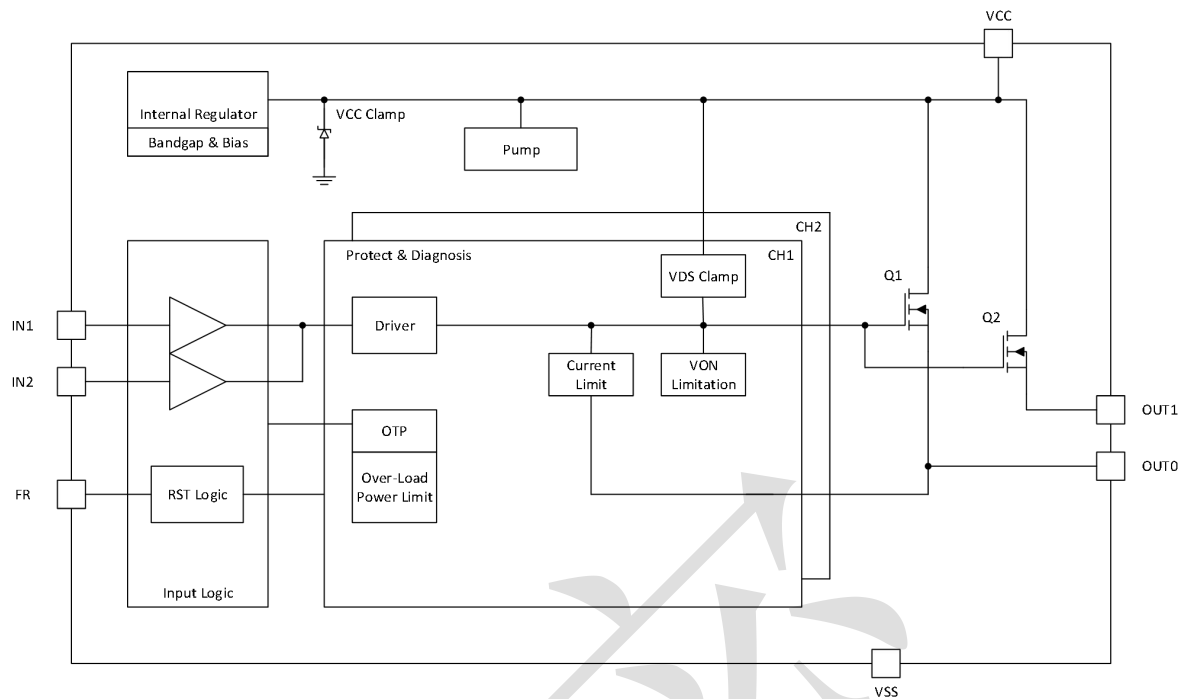
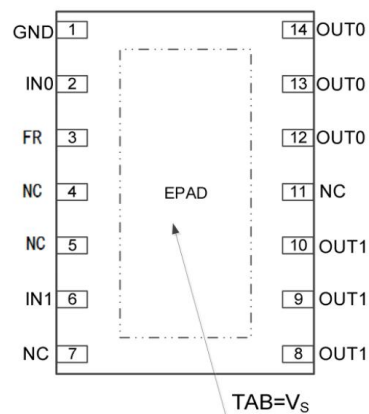


Table 1. Pin definition

Pin No.	Name	Function
1	GND	Ground connection.
2	IN0	Logic Input of channel 1.
3	FR	In case of latch-off for overtemperature/overcurrent condition, a low pulse on the FR_Stby pin is needed to reset the channel. The device enters in standby mode if all inputs and the FR_Stby pin are low
4,5,7,11	NC	NC
6	IN1	Logic Input of channel 2.
8,9,10	Out1	Power output of channel 2.
12,13,14	Out0	Power output of channel 1.

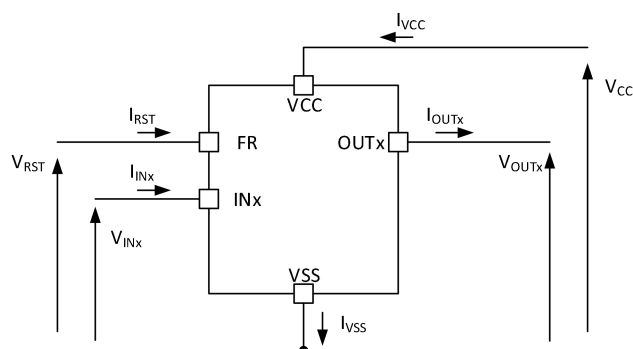
Figure 2. Configuration diagram of DFN9*6

引脚配置



Electrical specification

Figure 3. Current and voltage conventions



Absolute maximum ratings

Stressing the device above the ratings listed in the Table 3 may cause permanent damage to the device. These are stress ratings only and operation of the device at these or any other conditions above those indicated in the operating sections of this specification is not implied. Exposure to the conditions reported in this section for extended periods may affect device reliability.

Table 3. Absolute maximum ratings

Symbol	Parameter		Value	Unit
Vcc	DC supply voltage		52	V
-Vcc	Reverse DC supply voltage		0.3	V
-Ivss	DC reverse ground pin current		100	mA
IOUT	DC output current		Internally limited	A
-IOUT	Reverse DC output current		10	A
IIN	DC input current		-1 to 10	mA
IFR_Stby	Fault reset standby DC input current		-1 to 1.5	mA
VESD	Electrostatic discharge (human body model: R = 1.5 kΩ, C = 100 pF)	IN0, 1	4000	V
		OUT0, 1	4000	
		Vcc	4000	
VESD	Charge device model (CDM-AEC-Q100-011)		400	V

Thermal data

Symbol	Parameter	Value	Unit
TJ	Junction operating temperature	-40 to 150	°C
Tstg	Storage temperature	-55 to 150	°C

Electrical characteristics

8 V < V_{CC} < 36 V, -40 °C < T_J < 150 °C, unless otherwise specified.

Power Section

Symbol	Parameter	Test conditions	Min.	Typ.	Max.	Unit
V _{CC}	Operating supply voltage		8		36	V
V _{USD}	Undervoltage shutdown			6		V
V _{USDhyst}	Undervoltage shutdown hysteresis			0.5		V
R _{ON}	On-state resistance	I _{OUT} = 2 A, T _J = 25 °C		50		
		I _{OUT} = 2 A, T _J = 150 °C		100		mΩ
V _{clamp}	Clamp voltage	I _S = 20 mA		52		V
I _S	Supply current	Off-state, V _{CC} = 24 V, T _J = 25 °C, V _{IN} = V _{OUT} = 0 V, V _{FR_Stby} = 0 V		0.7		μA
		Standby state, V _{CC} = 24 V, T _J = 25 °C, V _{IN} = V _{OUT} = 0 V, V _{FR_Stby} = 5 V		1.5		mA
		On-state, V _{CC} = 24 V, V _{IN} = 5 V, I _{OUT} = 0 A		4		mA
I _{L(off)}	Off-state output current	V _{IN} = V _{OUT} = 0 V, V _{CC} = 24 V, T _J = 25 °C		0		
		V _{IN} = V _{OUT} = 0 V, V _{CC} = 24 V, T _J = 125 °C			10	μA

Switching characteristic

Symbol	Parameter	Test conditions	Min.	Typ.	Max.	Unit
t _{d(on)}	Turn-on delay time	R _L = 12 Ω		10		μs
t _{d(off)}	Turn-off delay time	R _L = 12 Ω		30		μs
(dV _{OUT} /dt) _(on)	Turn-on voltage slope	R _L = 12 Ω		0.75		V/μs
(dV _{OUT} /dt) _(off)	Turn-off voltage slope	R _L = 12 Ω		0.45		V/μs
W _{ON}	Switching energy losses during t _{won}	R _L = 12 Ω		0.6		mJ
W _{OFF}	Switching energy losses during t _{woff}	R _L = 12 Ω		0.7		mJ

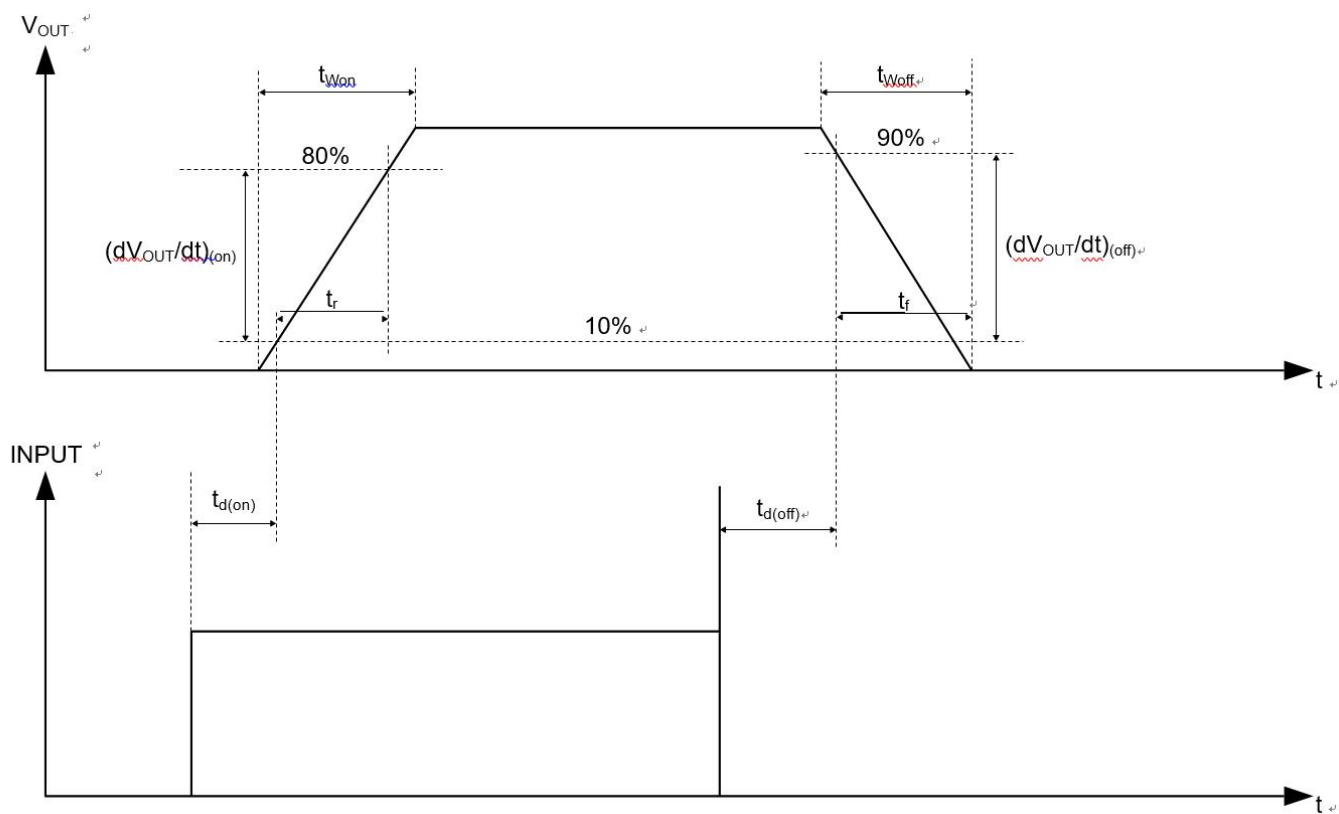
Logic Inputs

Symbol	Parameter	Test conditions	Min.	Typ.	Max.	Unit
V _{IL}	Input low level voltage				0.9	V
I _{IL}	Low level input current	V _{IN} = 0.9 V		2.2		μA
V _{IH}	Input high level voltage		2.1			V
I _{IH}	High level input current	V _{IN} = 2.1 V		4		μA
V _{I(hyst)}	Input hysteresis voltage			0.26		V
V _{ICL}	Input clamp voltage	I _{IN} = 1 mA		6.2		V
		I _{IN} = -1 mA		-0.7		V
V _{FR_Stby_L}	Fault_reset_standby low level voltage				0.9	V
I _{FR_Stby_L}	Low level fault_reset_standby current	V _{FR_Stby} = 0.9 V		2.2		μA
V _{FR_Stby_H}	Fault_reset_standby high level voltage		2.1			V
I _{FR_Stby_H}	High level fault_reset_standby current	V _{FR_Stby} = 2.1 V		5		μA
V _{FR_Stby(hyst)}	Fault_reset_standby hysteresis voltage			0.4		V
V _{FR_Stby_CL}	Fault_reset_standby clamp voltage	I _{FR_Stby} = 15 mA (t < 10 ms)		6.3		V
		I _{FR_Stby} = -1 mA		-0.7		V

Protections and diagnostics

Symbol	Parameter	Test conditions	Min.	Typ.	Max.	Unit
I _{limH}	DC short circuit current	V _{CC} = 24 V		29		
		5 V < V _{CC} < 36 V		20		A
I _{limL}	Short circuit current during thermal cycling	V _{CC} = 24 V, T _R < T _J < T _{TSD}		11		A
T _{TSD}	Shutdown temperature			155		°C
T _R	Reset temperature			145		°C
T _{HYST}	Thermal hysteresis (T _{TSD} - T _R)			8		°C

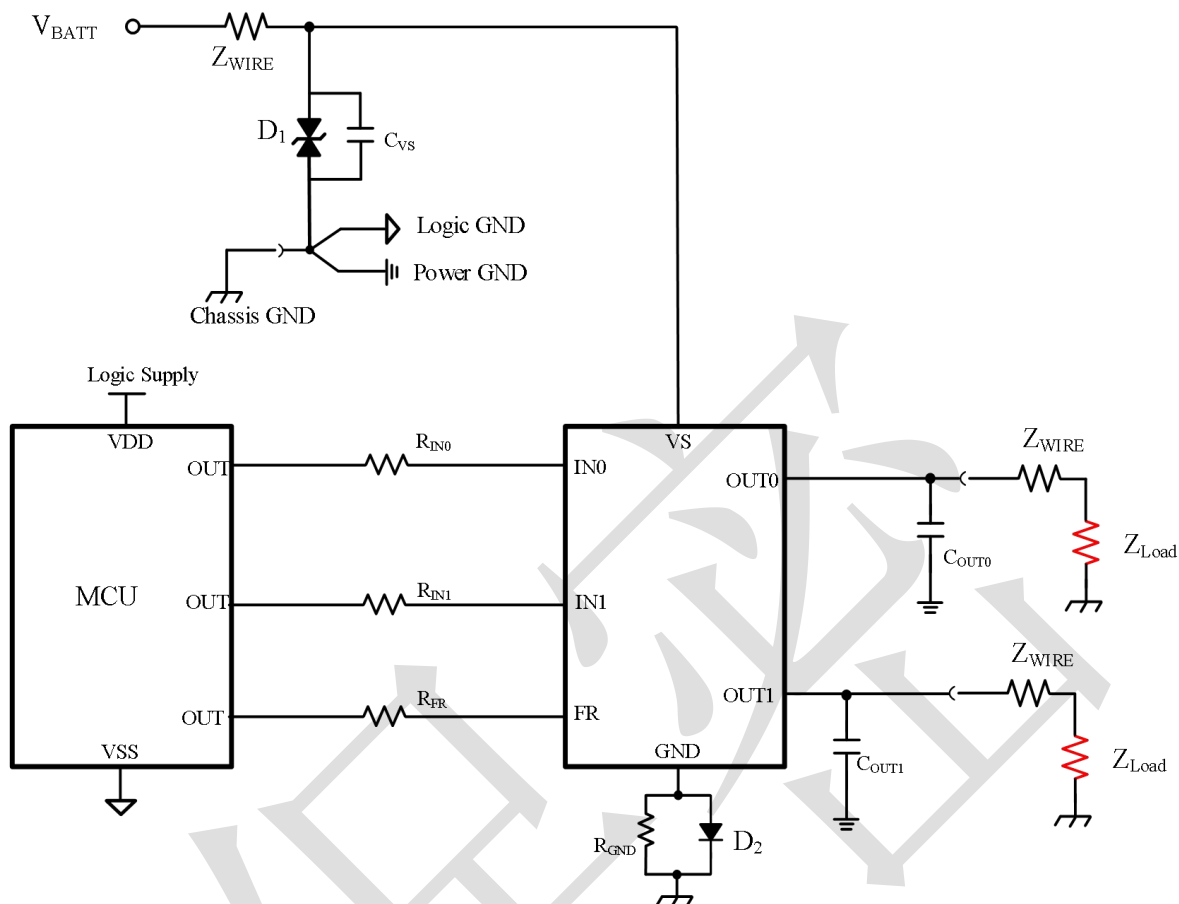
Figure 1. Switching characteristics



Electrical characteristics (curves)

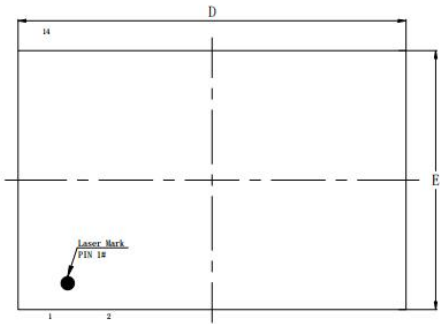
Application information

Figure 2. Application schematic

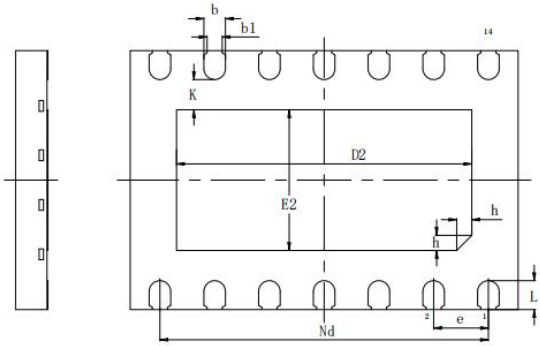


Reference	Value	Purpose
R_{IN0}, R_{IN1}	4.7k Ω	Protection of the microcontroller during Overvoltage and Reverse Polarity.
R_{FR}	4.7k Ω	Protection of the microcontroller during Overvoltage and Reverse Polarity.
R_{GND}	4.7 k Ω	Protection in case of Overvoltage and Loss of Battery.
C_{OUT}	10nF	Protection of output during ESD events and BCI
C_{VS}	100nF	Filtering of voltage spikes on the battery line
D_1	52V Zener diode	Protection of the device during overvoltage
D_2	1N4148	Protection of the device during reverse polarity

Package

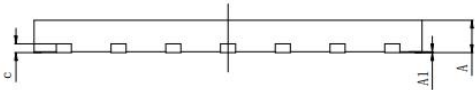


TOP VIEW



BOTTOM VIEW

SYMBOL	MILLIMETER		
	MIN	NOM	MAX
A	0.80	0.85	0.90
A1	0	0.02	0.05
b	0.45	0.50	0.55
b1	0.35REF		
c	0.203REF		
D	8.90	9.00	9.10
D2	6.75	6.85	6.95
e	1.27BSC		
Nd	7.62BSC		
E	5.90	6.00	6.10
E2	3.16	3.26	3.36
L	0.62	0.67	0.72
h	0.30	0.35	0.40
K	0.70REF		



SIDE VIEW

** 特殊设计: 无